SUPPORTING ADULT GUIDANCE

The second difference

Quadratics. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner meets the defining property of a quadratic sequence: a **constant second difference**. Building on Lesson 1 (where the first differences were seen to grow), this lesson takes the differences a **second time** and finds a constant. That constant is used three ways: as a **test** (constant second difference ⇒ quadratic; constant first difference ⇒ linear; neither ⇒ something else), as a **tool to continue** a sequence (the second difference gives the next first difference, which gives the next term), and as a **link to the rule** (for $y = a n^2$, the second difference is 2a). The organising object is the **difference table**: terms, first differences, second differences.

Driving Question: *How can one constant number reveal a hidden quadratic?*

The mathematics, in plain terms

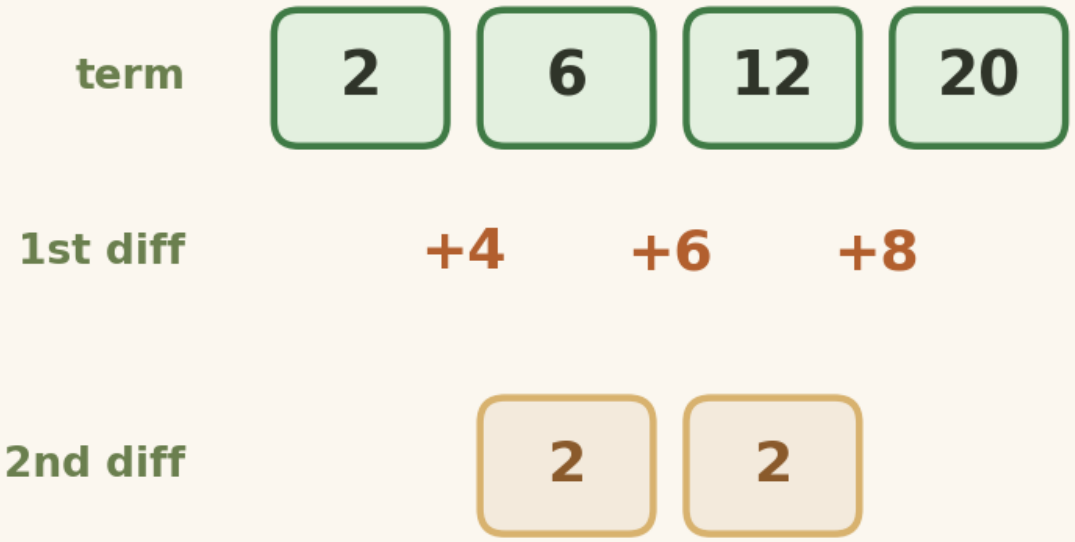
In Lesson 1 the learner wrote the gaps between terms — the **first differences** — underneath a sequence and saw them grow. This lesson does the same thing once more: write the differences **of those gaps** underneath, and for a quadratic they come out constant. Take 1, 4, 9, 16, 25: the first differences are 3, 5, 7, 9, and their differences — the **second differences** — are 2, 2, 2. That single repeating number is what makes the sequence quadratic. It gives a clean **test**: a constant first difference is linear (a straight-line pattern); a constant second difference is quadratic; if neither is constant, it is something else entirely (the interactive shows a doubling sequence, which is neither). The second difference is also a genuine **tool**. To continue 2, 6, 12, 20: the first differences are 4, 6, 8 and the second difference is 2, so the next gap is 8 + 2 = 10, and the next term is 20 + 10 = 30. Finally, there is a lovely link to the rule: for $y = a n^2$ the second difference is exactly **2a** — the plain square numbers (a = 1) have second difference 2, and doubling the rule to $2n^2$ doubles the second difference to 4. The whole lesson lives in the **difference table**, and the move to praise is a learner laying one out neatly and reading the bottom row.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — is it quadratic?

Build the difference table. If the second differences are all the same, it is a quadratic.

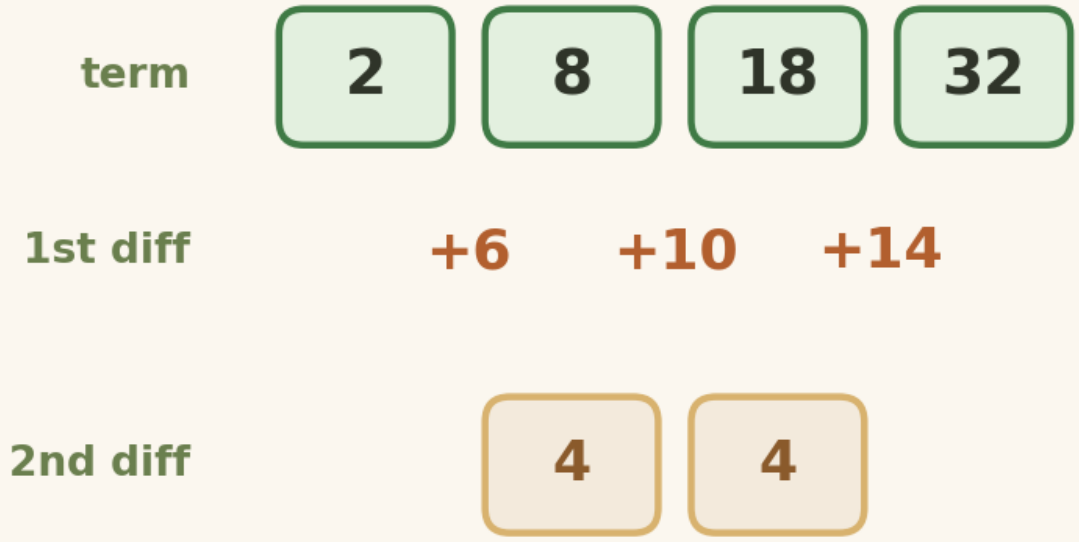


1. the first differences (the gaps)	4, 6, 8
2. the second differences (gaps of the gaps)	2, 2
3. the second difference is constant, so this is	quadratic

quadratic — the second difference is 2

Example 2 — continue using the second difference

The second difference is constant, so use it to reach the next term: it gives the next gap, and the gap gives the next term.



1. the first differences	6, 10, 14
2. the second difference (constant)	4
3. next gap = last gap + 4	14 + 4 = 18
4. next term = last term + next gap	32 + 18

the next term is 50

Read the bottom row of the table

Everything in this lesson lives in a **difference table**: the sequence on top, the first differences below, the second differences below that. The single organising habit is to **build the table and read its bottom row**. If the first-difference row is constant, the sequence is linear; if the second-difference row is constant, it is quadratic; if neither settles, it is neither. Keep the layout tidy and offset — each difference sits between the two numbers it came from — because the alignment is what makes the pattern visible. Two supports matter. First, insist the learner writes *both* difference rows even when they think they know the answer; the second row is the evidence. Second, when continuing a sequence, work **downwards then back up**: extend the constant second difference, add it to the last first difference to get the next gap, then add that to the last term. Resist introducing the nth-term formula; the link the learner should meet here is only the gentle one — that the second difference of a n^2 is 2a — which keeps the fingerprint tied to the rule without heavy algebra.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The fingerprint: reveal a difference table row by row until the constant second difference appears.	~10 min
Movement	Second-Difference Detective: use the constant second difference to continue a quadratic.	~12 min
Reflection	Is it quadratic?: sort sequences into linear, quadratic or neither using the difference table (dominant).	~14 min
Creativity	The second difference and the rule: see that y = a n^2 has second difference 2 × a.	~5 min
Rest	A pause after finding the steady thing beneath a pattern that would not stop changing.	~2 min
Nutrition	Mode B (intellectual): what the learning feeds — the idea of an invariant, a constant you can rely on and use.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Stopping at the first differences

Repair: “The first differences grow — that only tells you it is not linear. Take the differences again; the second row is the test.”

Thinking any increasing sequence is quadratic

Repair: “Check the second differences. If they are not constant it is not quadratic — the doubling sequence 1, 2, 4, 8 is neither.”

Confusing linear and quadratic

Repair: “Constant FIRST difference is linear. Constant SECOND difference is quadratic. Name which row is constant.”

Adding the second difference to the term instead of to the gap

Repair: “The second difference makes the next gap, not the next term. Next gap = last gap + second difference; then next term = last term + next gap.”

Thinking the second difference equals the number in front of n^2

Repair: “It is twice it. For $y = 3n^2$ the second difference is 2 × 3 = 6, not 3.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: first differences (the gaps between terms, from Lesson 1), squaring a number, and substituting a value into a rule such as $2n^2$. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner building and reading the difference table:

- “Write the first differences. Now write the differences of those — what is the bottom row?”
- “Is the bottom row constant? Constant first row means linear; constant second row means quadratic.”
- “The second difference is 2. What is the next gap? So what is the next term?”
- “For $y = 2n^2$, what will the second difference be? Check it against 2 × 2.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why take the differences twice?”

Because the first differences of a quadratic grow, so they do not reveal a constant on their own. Taking the differences again strips the growth away and leaves the constant that defines the quadratic.

“What if the second differences are not constant?”

Then it is neither linear nor quadratic — it is some other kind of sequence. The interactive includes a doubling sequence (1, 2, 4, 8, 16) whose second differences keep growing, to show this honestly.

“Do they need the nth-term formula?”

Not here. The only link to the rule in this lesson is the gentle one: for $y = a\,n^2$, the second difference is 2 × a. The full nth-term work comes later.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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